

SDR-Based Spectrum Monitoring System for FM Broadcast Compliance Assessment

Introduction

– *Purpose and Scope.* This document specifies the architecture, measurement framework, and processing algorithms for a Software-Defined Radio (SDR) based FM broadcast compliance monitoring system. It is intended for:

- RF engineers designing and deploying sensor nodes;
- Regulatory bodies evaluating technical adequacy and metrological traceability;
- Software developers implementing DSP processing pipelines;
- System operators responsible for calibration, validation, and evidence archiving.

This document is primarily a framework and requirements specification. The framework addresses the complete monitoring chain with explicit treatment of calibration hierarchy, uncertainty propagation, node health, multi-node coordination, and audit-ready reporting.

– *Regulatory Context.* The system targets compliance assessment under the following regulatory and technical frameworks:

- **FCC Part 73** (United States): Technical standards for FM broadcast service, including carrier frequency tolerance, occupied bandwidth, and emission mask requirements;
- **ANE Resolucin 105** (applicable jurisdiction): National regulatory provisions governing FM broadcast technical parameters;
- **ITU-R BS.412**: International technical standards for FM sound broadcasting;
- **ISO/IEC 17025**: General requirements for the competence of testing and calibration laboratories, applicable to measurement traceability and uncertainty evaluation.

For each deployment, the applicable regulatory basis shall be explicitly identified in the system configuration, measurement procedure, and reporting policy. All reported measurands, decision outcomes, and compliance determinations shall be traceable to the governing rule set, the calibration state of the measurement chain, and the documented processing and decision procedure used to produce the result.¹

¹Throughout this document, the key words “shall”, “shall not”, “should”, “should not”, and “may” are to be interpreted as described in RFC 2119 [?]. Requirements denoted by “shall” are mandatory for compliance with this specification.

1. System-Level Requirements for SDR-Based Monitoring Architectures

- *Reconfigurability and Flexibility:* The SDR platform shall support changes in monitoring objectives, signal-processing parameters, and compliance algorithms without hardware re-design. Functions such as demodulation, channelization, thresholding, occupancy analysis, and report generation shall be implemented in software to permit controlled adaptation to evolving regulatory requirements, extension to adjacent bands, and integration of validated processing methods. Any software or firmware change affecting measurement outputs shall be documented, version-controlled, and revalidated.
- *Cost and Deployability:* The monitoring node shall be economically deployable at scale while preserving the performance required for regulatory use. Low-cost SDR hardware and embedded processing platforms may be used, provided they do not compromise frequency accuracy, calibration validity, or operational reliability. The final architecture shall be justified on the basis of both technical adequacy and total deployment cost.
- *Remote Monitoring and Data Transport:* The system shall support remote transmission of measurement metadata, system-health information, and authorized derived products to a centralized server through IP-based networks. The architecture shall support continuous or scheduled delivery, local buffering during link outages, and reliable upload upon reconnection. Data products, transport rates, retention policies, and latency constraints shall be defined according to the monitoring mission, legal requirements, and available network capacity.
- *Cybersecurity and Access Control:* Data in transit shall be protected using authenticated encryption, and stored records shall be protected according to the system security policy. Administrative and operator access shall be controlled through authenticated and authorized mechanisms. Audit logs shall record configuration changes, software updates, user actions, and remotely triggered monitoring operations.
- *Multi-Channel Monitoring Capacity:* The DSP pipeline shall support real-time demodulation and compliance analysis of multiple FM broadcast channels within the instantaneous observation bandwidth and tuning constraints of the SDR front-end and processing platform. The minimum sustained concurrent channel count shall be established through benchmarking on the deployment hardware. Validation shall document processor utilization, memory consumption, storage throughput, end-to-end latency, and the maximum sustainable channel count without sample loss or measurement degradation.
- *Measurement Integrity, Calibration, and Traceability:* The monitoring architecture shall preserve the metrological integrity of all reported measurands. The receive chain, including antenna, RF front-end, SDR, timing reference, and processing stages, shall support calibration, periodic verification, documented uncertainty, and reproducible performance across deployed nodes. All reported measurands, compliance decisions, alerts, and archived evidence shall be traceable to the acquisition record, receiver configuration, timing reference, calibration state, processing configuration, uncertainty characterization, and applicable regulatory rule. Any change affecting the measurement chain shall require revalidation of the corresponding calibration and reporting functions.

To support this requirement, the system shall implement a *hierarchical calibration strategy* with three tiers, aligned with the regulatory significance of the measurement.

- *Tier 1: Primary (Laboratory) Calibration.* Each monitoring node shall undergo initial end-to-end calibration in a controlled laboratory environment, traceable to national or international standards. This calibration shall establish the absolute accuracy of the measurement reference plane, typically the antenna port, and shall characterize frequency response, power accuracy, linearity, and relevant environmental sensitivities. A unique calibration certificate and electronic calibration record shall be maintained for each node.
- *Tier 2: Secondary (Field) Verification.* Each node shall undergo periodic in situ verification to confirm that performance remains within acceptable limits relative to the primary calibration state. This procedure may use an internal reference source or a stable external signal. It need not re-establish absolute traceability, but shall provide a reliable check against predefined warning and alarm limits. Failure shall trigger a maintenance alert and suspend compliance use until resolved.
- *Tier 3: Relative (Channel) Calibration.* For measurements that do not require absolute power accuracy, a relative calibration across the observation bandwidth shall be maintained to control gain and phase distortion in spectral measurements. This calibration may be implemented through equalization filters, pilot-based methods, or passband characterization established during primary calibration.

2. Functional Decomposition of the Monitoring System

The architecture resulting from these design considerations is summarized in Figure 1, which identifies the principal hardware and software components and their interconnections.

- *RF Front-End (Hardware Layer):* The monitoring node employs a SDR receiver as the antenna-coupled RF front end for operation in the VHF-II broadcast band (87.5–108 MHz). This stage captures incident RF energy, tunes the band of interest, and produces digitized complex baseband samples for downstream processing. A VHF-II antenna interfaces to the SDR input, and an LNA may be added when required to compensate for feedline loss or weak received signals. The suitability of this front end is governed by instantaneous bandwidth, dynamic range, frequency-reference accuracy and stability, noise figure, and gain configuration, since these parameters bound the fidelity of derived compliance measurands.
- *Signal Acquisition and Digitization:* The SDR streams raw complex baseband (I/Q) samples to the embedded host processor through a high-speed USB interface. These samples represent the instantaneous tuned observation bandwidth and permit simultaneous inspection of multiple contiguous FM broadcast channels within a single acquisition window, reducing the need for sequential retuning. The selected sample rate shall be validated on the target platform to ensure sustained operation without dropped samples, taking into account USB throughput, host buffering, and processing load.
- *Software Processing Engine:* The software processing engine implements the end-to-end DSP pipeline for *FM broadcast compliance monitoring*. Its functions may include spectrum estimation, channelization, carrier detection, demodulation, occupancy assessment, feature extraction, alarm generation, and compliance-metric evaluation. The primary output of this layer is not user-traffic recovery, but reported measurands, decision outcomes, and supporting evidence suitable for regulatory review.

SDR Spectrum Sensing System

For FM Broadcasting Monitoring

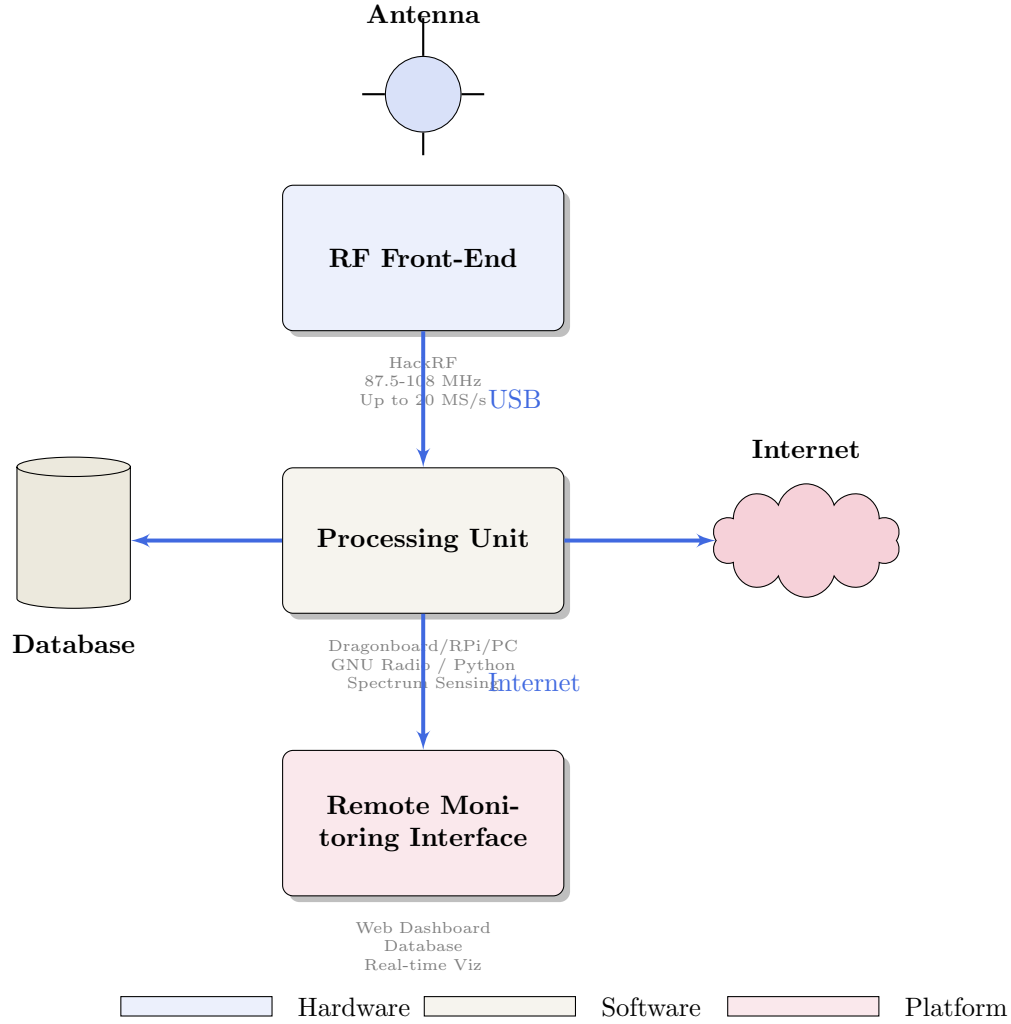


Figure 1: Core Architectural Components

- *Data Management and Reporting Layer*: Processed outputs shall be stored in a structured back-end using time-stamped records and associated metadata, including frequency, site, station identifier, measurement type, and compliance status. This layer supports historical trend analysis, event review, audit-ready report generation, controlled remote access through an API and monitoring dashboard, and automated notifications when predefined thresholds are exceeded in accordance with the operational monitoring policy.

3. FM Compliance Measurands and Traceability Requirements

For SDR-based FM broadcast monitoring, a *measurand* is a defined physical or derived quantity reported by the monitoring system for regulatory assessment. The architecture shall distinguish between *primary compliance measurands*, which directly support conformance evaluation, and *secondary observables*, which provide diagnostic or evidentiary support but do not by themselves establish compliance.

- *Primary Compliance Measurands*: Primary compliance measurands are those used directly to assess technical conformity under the applicable regulatory framework. They may include:
 - carrier frequency or frequency offset relative to the assigned channel (carrier frequency error: $\pm(0.5 + 2 \times f \times 10e^{-5})$ kHz);
 - calibrated received power or spectral power at a defined reference plane within the observation bandwidth (± 2.5 dB);
 - field strength, when supported by antenna-factor characterization and end-to-end calibration of the receive chain (± 3.5 dB);
 - occupied bandwidth and related spectral-containment indicators;
 - adjacent-channel leakage or out-of-band emission level;
 - modulation-related quantities, including peak deviation and, where required, multiplex power;
 - channel occupancy over time, including persistence metrics derived from a documented detection rule.
- *Secondary Observables*: Secondary observables support interpretation, diagnosis, and audit review. They shall not be treated as standalone compliance determinations unless explicitly required by the governing measurement procedure. They may include:
 - demodulated audio excerpts;
 - baseband or multiplex spectrum snapshots;
 - spectrograms and time-frequency visualizations;
 - anomaly scores or classifier outputs produced by statistical or machine-learning methods.
- *Measurement Conditions, Metadata, and Traceability*: Each reported measurand shall be stored together with its measurement conditions and associated metadata, including observation bandwidth, center frequency, sample rate, receiver gain state, antenna configuration, reference plane, calibration state, calibration record identifier, site identifier, timestamp reference, and processing-software version. For derived quantities, the processing method, averaging interval, uncertainty characterization, decision rule, and applicable regulatory threshold shall also be recorded to support verification, reproducibility, inter-node comparison, and technical review.

Table 1 defines the principal compliance measurands used by the SDR-based FM monitoring system. For each quantity, it specifies the operational definition, units, estimation method, calibration dependence, reporting cadence, and applicable compliance reference.

* Carrier-frequency error shall be reported as a compliance measurand only when the node frequency reference has been validated by a traceable calibration procedure or by an external disciplined reference.

Measurand	Symbol	Definition	Units	Estimation method	Calibration dependency	Reporting interval	Compliance reference
Carrier frequency error	Δf_c	Difference between the estimated channel center frequency and the assigned frequency of the monitored FM station.	Hz	Spectral peak, centroid, or model-based estimation referenced to the node frequency standard.	High; depends on frequency-reference accuracy, offset calibration, and drift compensation (*).	Periodic and event-triggered	Assigned-frequency tolerance specified by the applicable regulator or station license.
Calibrated received power	$P_{rx}(B)$	Power measured at a defined receiver reference plane over a stated bandwidth B after gain-state and frequency-response correction.	dBm or dB μ V	PSD estimation followed by integration over the defined analysis bandwidth after receiver calibration.	High; requires gain calibration, front-end response correction, and a documented reference plane.	Periodic	Site- or policy-specific threshold; suitable for compliance use only when the reference plane is defined.
Field strength	E	Electric field strength at the monitoring antenna location, inferred from the calibrated receive chain and antenna factor.	dB μ V/m	Conversion from calibrated received level using antenna factor, cable loss, and front-end corrections.	Very high; valid only when antenna factor and end-to-end chain calibration are available.	Periodic	Applicable field-strength rule, coordination criterion, or service-limit criterion adopted by the jurisdiction.
Occupied bandwidth	B_{occ}	Width of the frequency band containing a specified percentage of the total integrated mean emission power.	kHz	Direct $\beta\%$ occupied-bandwidth measurement on the recorded spectrum.	Moderate; depends on frequency-response flatness, RBW/FFT configuration, and interference control within the analysis span.	Periodic	Occupied-bandwidth or spectral-containment rule defined by the governing standard or license condition.
Adjacent-channel out-of-band emission level	L_{adj} / L_{OOB}	Emission level measured in adjacent-channel or out-of-band regions, reported either relative to the in-band reference level or at the calibrated receiver reference plane.	dBc or dBm	Spectral integration or peak-envelope evaluation in predefined offset bands relative to the assigned carrier.	High; requires dynamic-range validation, front-end linearity, and spectral calibration.	Periodic and event-triggered	Emission-mask or attenuation rule adopted by the regulator, evaluated for the stated offset region and reference level.
Peak deviation / multiplex-related indicator	$\Delta f_{max}, P_{MPX}$	Maximum instantaneous frequency deviation and, where required, a multiplex-related modulation quantity derived from the FM baseband or demodulated composite signal.	kHz, dBc, or other adopted MPX metric	FM demodulation followed by deviation estimation and multiplex-power measurement using the adopted method.	High; depends on demodulator linearity, deviation calibration, and composite baseband conditioning.	Windowed estimate with periodic summary	Applicable deviation or multiplex-power rule under the adopted monitoring procedure.
Channel occupancy over time	O_T	Fraction of the observation interval during which the channel is declared occupied according to a documented detection rule.	%	Threshold-, energy-, or feature-based detection over repeated observation windows with a defined revisit strategy.	Moderate; depends on threshold calibration, noise-floor estimation, and decision logic.	Rolling summary	Occupancy definition and decision rule established by the monitoring policy.

Table 1: Operational definition of principal FM compliance measurands. Here, $\beta\%$ occupied bandwidth denotes the minimum frequency interval containing $\beta\%$ of the total integrated mean emission power.

Classification Framework:

The SDR monitoring system shall classify each measurand based on hardware-constrained measurement quality. This classification dictates permissible regulatory use and defines required validation, calibration, and uncertainty characterization.

Three capability classes are defined:

- *Compliance-Grade*. The measurand may be used directly to support formal regulatory compliance determinations, provided that all calibration, validation, and uncertainty requirements specified in this document are satisfied. The hardware platform, processing implementation, and end-to-end measurement chain have been validated to meet the applicable regulatory tolerance and uncertainty limits.
- *Screening-Grade*. The measurand may be used for anomaly detection, event screening, situational awareness, or preliminary assessment, but it shall not be used as the sole basis for a formal non-compliance determination without corroboration from a compliance-grade measurement or independent validation. Screening-grade measurements may trigger alerts, operator review, or deployment of higher-capability instrumentation, but they do not by themselves constitute regulatory evidence.
- *Unsupported*. The measurand cannot be reliably estimated with the deployed hardware configuration and shall not be reported for either compliance or screening use. Attempting to measure an unsupported quantity may produce outputs that are numerically valid but metrologically meaningless or systematically biased beyond correction.
- *Conditional Compliance-Grade*. The measurand may reach compliance-grade status on this hardware platform, but only after all enumerated preconditions in the Enhancement column of Table 3 have been implemented, independently validated, and recorded in a calibration certificate. Prior to that validation, the measurand shall be treated as Screening-Grade for all regulatory purposes. The preconditions are mandatory, not recommended.²

Metadata Requirements for Capability-Dependent Reporting

Every reported measurand shall include in its metadata record:

- The capability class under which it was acquired (compliance-grade, screening-grade, unsupported),
- Hardware platform identifier and configuration (ADC resolution, gain state, reference source),
- Applicable calibration tier(s) and calibration record identifier,
- Measurement uncertainty or, for screening-grade results, an explicit uncertainty disclaimer,
- Any hardware augmentation applied (external GPSDO, LNA, filter),
- Software version and processing-method identifier.

Screening-grade measurands shall be flagged in all reports and databases as `FOR_SCREENING_ONLY` and shall not appear in formal compliance summaries without explicit override justification and independent validation.

²While the above standard classes exist for professional equipment, this ad hoc class is suitable for low-cost SDR boards with several limitations.

Uncertainty Budget and Reporting

All measurements intended for regulatory compliance use shall be accompanied by a documented uncertainty evaluation in accordance with the Guide to the Expression of Uncertainty in Measurement (GUM) [?]. The uncertainty budget shall account for all known sources of random and systematic error, shall be traceable to the calibration hierarchy defined above, and shall be reported alongside the measurand value in all compliance records and evidence archives.

Uncertainty Classification, the associated Uncertainty-Aware Compliance Decisions, and Periodic Re-Evaluation Requirements are detailed in Appendix

4. Node-Level DSP Pipeline for FM Compliance Monitoring

The compliance monitoring pipeline transforms raw complex baseband observations into reported measurands, decision outcomes, and supporting evidence. However, the technical rigor of such a pipeline implementation is often challenged by the hardware-software interface.

The principal processing stages are shown in Fig. 2 and described below.



Figure 2: Implemented SDR-based FM broadcast compliance monitoring pipeline. Raw I/Q samples are processed through six stages, from wideband acquisition and conditioning to reporting and evidence archiving, to enable automated regulatory assessment.

- *Wideband Acquisition & Conditioning:* Raw I/Q samples are acquired from the SDR front end together with receiver-configuration and timing metadata. This stage establishes the wideband observation used by all downstream analyses.

Its principal functions are:

- *Hardware configuration:* initialization of SDR operating parameters, including center frequency, sample rate, gain state, and front-end amplification settings;
- *Sample streaming:* transfer of raw complex baseband samples from the SDR device to the embedded host processor through the USB interface;
- *Buffer management:* temporary storage and flow control to reduce the risk of sample loss during sustained acquisition;
- *Metadata association:* tagging of captured data with timing, configuration, and calibration-state information;
- *Signal conditioning:* preliminary preparation of the wideband observation, including DC-offset mitigation, optional digital frequency translation, and other conditioning steps required before channelization and spectral analysis.

- *Channelization, Isolation & Tracking:* The conditioned wideband observation is analyzed to detect active FM broadcast channels within the monitored span. Detected channels are isolated and tracked over time to support stable downstream measurement.

Its principal functions are:

- *Wideband spectral survey*: estimation of spectral content over the monitored band to identify candidate FM emissions and their approximate center frequencies and occupied regions;
 - *Channel detection*: application of detection criteria to distinguish active broadcast channels from noise, interference, or spurious responses;
 - *Channel isolation*: extraction of individual FM channels through digital channelization, filtering, and frequency selection;
 - *Frequency tracking*: monitoring of the spectral position of each detected channel to compensate for drift, tuning offsets, or minor carrier variations;
 - *Persistence assessment*: evaluation of channel continuity and temporal presence to support occupancy estimation, event detection, and stable downstream demodulation.
- *Optional Per-Channel Demodulation & Identification*: Each isolated FM channel is demodulated to recover the signal products required for modulation analysis. Auxiliary identification functions may be used to support station recognition, event attribution, or diagnostic interpretation, but they do not constitute primary compliance criteria.

Its principal functions are:

- *FM demodulation*: conversion of each isolated RF channel into its corresponding baseband representation for subsequent compliance analysis;
 - *Composite baseband recovery*: extraction of the FM multiplex signal components relevant to deviation analysis, pilot detection, stereo-related features, and other modulation-dependent measurements;
 - *Baseband conditioning*: preparation of the recovered signal through filtering, normalization, and related operations required by the adopted estimation methods;
 - *Auxiliary station identification*: use of identifying signal characteristics, such as spectral signatures, pilot structure, metadata cues, or program-dependent features, to support station recognition and event attribution;
 - *Secondary feature classification*: optional classification of non-primary signal attributes for diagnostic interpretation, anomaly screening, or evidentiary support.
- *Compliance Measurand Estimation*: For each monitored channel, the system computes the operationally defined measurands required for regulatory assessment. Its main functions are:
- *Carrier-frequency estimation*: determination of channel center frequency and its deviation relative to the assigned broadcast frequency, referenced to the adopted frequency standard and calibration state;
 - *Power-related estimation*: computation of calibrated received power, spectral power, or other amplitude-related quantities at the defined reference plane, subject to the applicable gain and frequency-response corrections;
 - *Bandwidth and spectral-containment estimation*: evaluation of occupied bandwidth and related spectral-containment indicators from the recorded spectrum using the adopted analysis bandwidth and decision rule;
 - *Adjacent- and out-of-band emission analysis*: measurement of spectral leakage or emission level in adjacent-channel and out-of-band regions relative to the in-band reference level or other stated compliance reference;

- *Modulation-related estimation*: extraction of deviation-based and multiplex-related quantities from the demodulated baseband or composite signal;
 - *Occupancy estimation*: determination of channel occupancy over time using the documented detection rule, observation window, and revisit strategy;
 - *Result association*: association of each reported measurand with the relevant processing configuration, averaging interval, calibration state, and metadata.
- *Decision Logic & Alerting*: The estimated measurands are compared against applicable regulatory limits, license conditions, or monitoring rules. When threshold exceedances or anomalous conditions are detected, the system generates alerts and records the supporting decision context.

Its principal functions are:

- *Compliance-rule matching*: comparison of each estimated measurand against the applicable thresholds, license requirements, or monitoring criteria;
- *Decision logic execution*: application of the adopted decision rules to classify the observed channel state as compliant, non-compliant, uncertain, or requiring further review;
- *Threshold exceedance detection*: identification of events in which one or more measurands depart from their permitted range;
- *Anomaly handling*: processing of abnormal system or signal conditions that merit operator attention even when a formal violation is not established;
- *Alert generation*: creation of notifications, warning flags, or event markers for operator review;
- *Decision-context recording*: storage of supporting measurands, timestamps, processing parameters, rule set, and relevant metadata required to justify the alert or compliance decision.

The rule evaluation stage shall compare each estimated measurand x against the applicable regulatory threshold T using an explicitly defined decision rule that accounts for measurement uncertainty U . The decision logic depends on the regulatory framework, risk tolerance, and adopted compliance philosophy. Three supported decision frameworks are detailed in Appendix.

- *Reporting & Evidence Archiving*: The final stage stores reported results, selected signal products, and decision records in a structured backend for later retrieval, review, and audit.

Its principal functions are:

- *Measurement persistence*: storage of reported measurands, decision outcomes, and associated metadata in a structured backend;
- *Evidence preservation*: retention of selected supporting artifacts, such as PSD estimates, spectrograms, demodulated excerpts, and event-linked channel records;
- *Compliance reporting*: generation of summaries, station-level assessment records, and historical trend outputs suitable for operational monitoring and formal review;
- *Alert-log management*: archival of alert histories, event markers, severity levels, and operator-facing notifications;

- *Controlled access and retrieval:* provision of structured mechanisms for querying, visualizing, and exporting stored results in accordance with the monitoring policy and access-control model.

The estimation background of the processing pipeline is specified for each Compliance Measurement and in Appendix ??.

5. Array-Level Coordination for Distributed Regulatory Monitoring

In a distributed regulatory monitoring deployment, the sensor array shall be treated as a coordinated measurement system rather than as a collection of independent SDR nodes. Each node may acquire, preprocess, and estimate local measurands independently, but any array-level conclusion shall be formed only after the participating nodes have been brought into a common calibration, timing, reference, and health framework. This layer is necessary to ensure that multi-node conclusions remain technically comparable, metrologically defensible, and auditable under regulatory review.

- *Inter-node calibration.* Each node shall retain its own primary, field, and relative calibration records, but the array shall additionally maintain an inter-node calibration model that quantifies residual differences between nodes for the measurands intended for joint use. This model shall include, at minimum, frequency-reference offset, gain bias, passband-shape deviation, timestamp offset, and any site-dependent correction needed to compare measurements across locations. Inter-node calibration shall be verified periodically using a shared reference signal, a transportable reference source, or a controlled over-the-air reference transmission. Array-level fusion shall use only nodes whose residual mismatch remains within predefined acceptance limits for the relevant measurand.
- *Timing alignment.* All nodes contributing to common event detection, occupancy estimation, or corroborated compliance decisions shall operate against a common time base traceable to UTC or another approved reference. Timestamp uncertainty, holdover behavior, synchronization-loss state, and timing-validation results shall be recorded in metadata. For array processing, observations shall be aligned to a documented common epoch and window definition, with explicit tolerance on allowable inter-node skew. If the required timing tolerance is exceeded, the affected node may continue local monitoring, but it shall be excluded from time-critical multi-node conclusions until synchronization is restored.
- *Common reference distribution.* The array shall distribute or derive a common frequency and time reference appropriate to the target measurement class. For carrier-frequency compliance, the use of a disciplined external reference shall be preferred where the node-local oscillator is otherwise insufficiently stable for the required tolerance. The reference architecture shall specify the reference source, distribution method, holdover strategy, fault detection, and the procedure followed when a node loses reference lock. Any measurement obtained during reference loss shall be flagged according to its impact on metrological validity and may be downgraded to screening-only use where required.
- *Sensor health coordination.* The array management layer shall continuously supervise node health using both hardware and measurement-quality indicators. These shall include SDR connectivity, sample-loss status, gain state, clipping, DC contamination, I/Q residuals, thermal

alarms, storage availability, network reachability, calibration validity, synchronization state, and reference-lock state. Health supervision shall classify each node as at least one of the following: operational for compliance use, operational for screening only, degraded, or unavailable. Array-level conclusions shall only use nodes whose health state is compatible with the uncertainty and decision rule of the associated measurand.

- *Cross-node consistency checks.* When multiple nodes observe the same licensed emission or event, the system shall perform cross-node consistency analysis before issuing a fused regulatory conclusion. The consistency checks shall compare overlapping measurands after correcting for known site, antenna, and propagation differences. At minimum, the framework should evaluate consistency of carrier-frequency estimate, event time, occupancy declaration, and any jointly used spectral or modulation indicator. Persistent disagreement beyond the expected uncertainty budget shall trigger one of three outcomes: exclusion of the outlying node, classification of the event as uncertain, or escalation for operator review. Cross-node agreement shall strengthen confidence, but disagreement shall never be silently averaged away.
- *Contribution of multiple nodes to one regulatory conclusion.* The framework shall define a formal fusion rule for converting node-level results into an array-level conclusion. That rule shall specify:
 1. eligibility criteria for participating nodes;
 2. the measurands that may be fused and those that shall remain node-local;
 3. weighting factors based on calibration state, uncertainty, health, and geometry;
 4. the required corroboration level for declaring a non-compliance event; and
 5. how inconclusive or conflicting evidence is handled.

For example, the array may adopt a rule in which a regulatory alert is issued only when at least two eligible nodes independently report the same exceedance within a defined time window and with compatible uncertainty bounds, while a single-node exceedance is retained as a provisional alert pending corroboration or review. The exact fusion rule shall depend on the measurand: carrier-frequency anomalies may be corroborated by agreement in calibrated frequency error, whereas occupancy or interference events may rely on synchronized temporal coincidence across nodes.

- *Array-level traceability and evidence.* Every fused decision shall preserve the full chain of evidence linking the array conclusion to the underlying node records. The archived record shall identify the participating nodes, their calibration states, synchronization states, health states, local uncertainties, fusion weights, consistency-check outcomes, and the decision-rule version used to form the final conclusion. This ensures that the array result remains auditable and can be reconstructed during technical or legal review.
- *Degraded-mode operation.* The framework shall explicitly distinguish between full array operation and degraded array operation. Loss of one node, loss of common reference, excessive timing skew, or calibration expiry shall not necessarily halt all monitoring, but it shall automatically restrict which measurement classes remain valid. In degraded mode, the system may continue screening, situational awareness, or evidence collection, while suspending array-level compliance determinations that depend on the missing capability.

Baseline HackRF One Platform Assessment

Measurement Capability, Platform Limitations, and Suitability

The HackRF One was selected for large-scale deployment on the basis of cost and reconfigurability. Its use as a primary sensor node entails a trade-off between cost-efficiency and metrological rigour that must be explicitly characterised before any compliance determination is made.

Regulatory standards (e.g., FCC Part 73, ANE Resolucin 105) often mandate out-of-band emission attenuation exceeding 80 dB, whereas the HackRF One’s 8-bit ADC offers a theoretical dynamic range of only ≈ 48 dB ($6.02 \times 8 + 1.76$ dB). In regulatory monitoring, the Effective Number of Bits (ENOB) is more critical than nominal bit depth: thermal noise, non-linearity, and input-power-dependent compression typically reduce practical resolution to ≈ 7 bits (≈ 42 dB), with further degradation at the band edges of the 20 MHz observation window. This limitation renders ACLR measurements and any measurand dependent on high dynamic range unreliable without external filtering and strict gain-staging.

To maintain metrological integrity, strict gain-staging protocols are required. Peak signal levels shall be maintained at least 6 dB below full scale to prevent clipping. In distributed deployments where the signal environment varies across nodes, this requires either manual gain profiling at each site or verified AGC with mandatory state logging. If AGC is employed, its state **shall** be recorded and timestamped for every sample block; measurements acquired without this metadata **shall not** be used for calibrated received power reporting and shall be flagged as invalid in the acquisition record.

The hardware’s wideband nature introduces three DSP-level pathologies that affect different parts of the compliance measurand set:

- **DC spectral artifact.** The LO leakage produces a narrowband spike at DC in the baseband spectrum. Offset tuning (setting $f_{\text{LO}} = f_{\text{target}} + f_{\text{off}}$, $f_{\text{off}} = f_s/4$) displaces the spike away from the measurement band and is the required mitigation. This increases acquisition complexity: precise mapping between the LO frequency, applied offset, and true RF frequency must be maintained and recorded in metadata.
- **I/Q imbalance.** Amplitude imbalance g and phase error φ between the I and Q channels produce a mirror image of the desired signal at $-f_{\text{offset}}$ in the baseband spectrum. The correction applied in the preprocessing stage (a 2×2 widely-linear matrix with scalar g and φ) is frequency-independent. Across the 20 MHz observation bandwidth, the HackRF One’s imbalance parameters are frequency-dependent; the scalar correction leaves a residual imbalance at frequencies remote from the calibration tone. Nodes operating with channel targets more than approximately 5 MHz from the calibration tone frequency **shall** re-estimate g and φ at a tone within 500 kHz of the target channel, or shall bound the residual mirror power by measurement and record it as an uncertainty contributor. The corrected mirror power **shall** remain below -40 dBc after correction; results exceeding this threshold shall trigger the `IQ_RESIDUAL` flag and shall not be used for occupied-bandwidth compliance estimation.
- **Frequency-dependent noise floor.** The receive chain exhibits a non-flat noise and gain profile across the observation bandwidth. A relative (Tier 3) calibration is required to characterise and correct this variation. The detection margin Δ_{det} used in Stage 4 channel detection is not a fixed absolute value; it **shall** be set as a site-calibrated offset above the locally verified noise floor, derived from the Tier 2 noise-floor verification required by the channel

occupancy preconditions in Table 3. A default value of 8 dB may be used only during initial commissioning prior to site verification; it **shall not** be used for compliance-grade or conditional-compliance-grade occupancy reporting.

Local-oscillator phase noise and the nominal ± 20 ppm crystal reference accuracy constrain frequency stability. For carrier-frequency compliance, an external GPS-Disciplined Oscillator (GPSDO) providing a traceable 10 MHz reference is the required mitigation; validated post-hoc calibration alone is insufficient because it cannot correct for in-situ thermal drift between calibration intervals. Any measurement of carrier frequency error acquired without a locked GPSDO or equivalent disciplined reference **shall** be classified as screening-grade and flagged `UNCALIBRATED_REFERENCE` regardless of any other configuration.

Additionally, the Stage 4 harmonic-rejection test is currently implemented as a placeholder and performs no rejection. Until this function is implemented and validated, nodes **shall not** be declared suitable for compliance-grade or conditional-compliance-grade use in deployment environments where intermodulation products from co-channel or adjacent FM stations may produce spurious candidates within the monitored span. Operators shall assess intermodulation risk at each site prior to compliance-use declaration by examining the wideband PSD for candidate energy at harmonic offsets from strong in-band carriers.

If any measurand class exceeds the validated hardware capability as defined in Table 3, the node **shall** be declared unsuitable for that measurand and the corresponding output **shall** be suppressed or flagged `UNSUPPORTED` in all reports.

Upgrade Paths and Hybrid Architectures

Deployments that require compliance-grade measurement of measurands currently classified as Screening-Grade or Unsupported in Table 3 shall implement one of the following strategies. The strategies differ in scope: Strategy 1 achieves compliance-grade coverage across all primary measurands; Strategy 2 achieves compliance-grade determination through array-level corroboration while retaining HackRF One screening nodes; Strategy 3 elevates a defined subset of measurands to compliance-grade or Conditional Compliance-Grade on the HackRF One platform through hardware augmentation and validation. In all cases, measurands classified as Conditional Compliance-Grade remain subject to the preconditions specified in Table 3, regardless of which strategy is adopted.

1. **Hardware Upgrade.** Replace the HackRF One with a higher-performance SDR platform offering 12–16 bit ADC resolution, lower phase noise, improved SFDR, and a factory-calibrated or externally disciplined frequency reference. Candidate platforms include the Ettus Research USRP B200/B210 and the Airspy R2 or Airspy Mini (24–1800 MHz); equivalent instrumentation-grade receivers operating across the VHF-II band (87.5–108 MHz) are also acceptable. Full Tier 1 calibration of the replacement platform is mandatory prior to compliance-use declaration. Measurands classified as Conditional Compliance-Grade in Table 3 remain subject to their precondition lists even after a hardware upgrade.
2. **Hybrid Screening and Validation Architecture.** Deploy HackRF One nodes for wide-area screening and event detection (Screening-Grade), supported by a smaller number of compliance-grade validation nodes equipped with higher-capability instrumentation at strategic locations. When a screening node triggers an alert, the event shall be corroborated by a compliance-grade node before any formal regulatory determination is issued. Corroboration shall follow the array-level fusion rules defined in Section 5, including eligibility criteria,

Table 2: Theoretical vs. practical hardware performance of the HackRF One for regulatory FM monitoring. Affected measurands are identified to make the table actionable.

Parameter	Theoretical	Practical (HackRF One)	Impact on compliance (affected measurands)
Dynamic range (ENOB-limited)	48.2 dB (8-bit)	35–45 dB (≈ 7 bit effective)	Insufficient for high-attenuation mask checks. Primary impact: ACLR, field strength.
SFDR ^a	≈ 48 dBFS ^a	50–55 dBc ^a	Limited ability to distinguish weak spurs near strong carriers. Primary impact: ACLR, occupied bandwidth near band edges.
I/Q imbalance (scalar correction residual)	0 dB mirror rejection (ideal)	< -40 dBc within ± 5 MHz of cal. tone; degrades at band edges	Mirror images bias spectral estimates and generate false candidates. Primary impact: carrier frequency estimation, occupied bandwidth. Frequency-dependent; re-calibration required per site channel (see Section).
LO phase noise	< -100 dBc/Hz at 10 kHz offset (ideal)	-80 to -85 dBc/Hz at 10 kHz offset (stock XO)	Broadens spectral estimates and raises close-in noise floor. Primary impact: carrier frequency error, occupied bandwidth edge interpolation. Mitigated by GPSDO with low-phase-noise 10 MHz reference.
Sampling rate (sustained)	Up to 20 Msps	Reliable at 10–16 Msps	Limits instantaneous observation bandwidth for multi-channel survey. Primary impact: wideband acquisition coverage, multi-channel occupancy.
Frequency reference accuracy	N/A (XO-dependent)	± 20 ppm (stock XO)	± 2 kHz absolute error at 100 MHz; exceeds carrier-frequency compliance tolerance without GPSDO. Primary impact: carrier frequency error.

^a Theoretical SFDR is expressed in dBFS (relative to full scale); practical SFDR is expressed in dBc (relative to a carrier tone). These are not directly comparable: a strong in-band carrier raises the apparent SFDR in dBc while the absolute spurious floor in dBm remains constrained by the 8-bit dynamic range. The dBFS figure governs compliance headroom calculations.

Sources: Dynamic range and SFDR values based on the MAX2837 datasheet and empirical HackRF One characterisation. Phase noise figures are indicative; site measurement is recommended for compliance-critical deployments.

inter-node calibration verification, timing alignment, and consistency checks. A single-node screening alert that cannot be corroborated within the defined time window and uncertainty bounds shall be retained as a provisional alert and escalated to operator review; it shall not constitute a standalone compliance determination.

3. **External Reference and Filtering Augmentation.** Retain the HackRF One platform and augment each node with the following, all of which are mandatory for compliance-use declaration of the affected measurands:

- External GPSDO providing a traceable 10 MHz reference and 1 PPS timing signal (required for carrier frequency error);
- External preselector or tunable bandpass filter to improve dynamic range for adjacent-channel measurements;
- Documented LNA with characterised gain and noise figure, inserted and calibrated as part of the Tier 1 receive-chain calibration;
- Automated AGC state logging and gain-step calibration, with per-block metadata recording as specified in Section ;
- Implementation and validation of the Stage 4 harmonic rejection function prior to compliance-use declaration in any environment where intermodulation products from co-channel or adjacent FM stations may produce spurious candidates.

This approach elevates carrier frequency error and channel occupancy toward their respective target capability classes (Screening-Grade and Conditional Compliance-Grade), and may elevate occupied bandwidth to Conditional Compliance-Grade, subject to completion of all preconditions listed in Table 3 and documented uncertainty characterisation under the GUM framework. Field strength and adjacent-channel emission remain Unsupported under this strategy; compliance-grade assessment of these measurands requires Strategy 1.

Capability Classification by Primary Measurand

Table 3 summarises the baseline capability classification for each primary compliance measurand when the monitoring node is implemented using the HackRF One SDR receiver. The classification reflects the limitations imposed by the 8-bit ADC (effective ≈ 7 bits, ≈ 42 dB dynamic range), ± 20 ppm uncalibrated frequency reference, front-end linearity, and noise floor characteristics described in Section .

The classification is a *baseline*: a node that has completed all mandatory preconditions listed for a Conditional Compliance-Grade measurand, and holds a current calibration certificate attesting to that completion, is reclassified to Compliance-Grade for that measurand only. In the absence of a valid certificate, the measurand shall revert to Screening-Grade regardless of any other system configuration. The Conditional Compliance-Grade class is defined in Section 3.

For Unsupported measurands, the listed factors are hardware-imposed disqualifying limitations that cannot be resolved by calibration or software alone; the required platform capability is stated to guide upgrade decisions. For Screening-Grade measurands, the listed items are mandatory qualifications that must be satisfied before the measurand output is used for any monitoring purpose.

Table 3: Measurement Capability Matrix for HackRF One-Based Monitoring Node. Classifications are baseline values; see paragraph above for reclassification conditions.

Measurand	Capability Class	Qualification, Preconditions, or Limiting Factors ^a
Carrier Frequency Error	Screening	External GPSDO required as the sole path to compliance-grade; post-hoc calibration alone is insufficient to correct in-situ thermal drift. Tier 1 calibration of frequency reference mandatory. Residual uncertainty shall be characterised and reported. Measurements acquired without a locked GPSDO shall be flagged <code>UNCALIBRATED_REFERENCE</code> and treated as screening-grade regardless of other configuration. ^b
Calibrated Received Power	Screening	AGC state shall be recorded and timestamped for every sample block; measurements acquired without this metadata shall not be used for power reporting. Tier 1 and Tier 2 calibration of gain chain mandatory. Absolute power accuracy limited by gain quantisation ($\pm 2-8$ dB steps) and front-end uncertainty. Valid only when clipping is absent and gain state is recorded.
Field Strength	Unsupported	Disqualifying limitations: (i) dynamic range of 35–45 dB is insufficient for weak-signal monitoring in the presence of strong co-channel or adjacent interference; (ii) no factory antenna-factor characterisation is available for the HackRF One receive chain. Reclassification requires a platform with ≥ 12 -bit ADC, full end-to-end Tier 1 calibration including antenna factor, feedline loss, and environmental correction, and a validated site-specific antenna pattern.
Occupied Bandwidth	Conditional Compliance	Mandatory preconditions — all must be satisfied and attested by calibration certificate: (1) FFT size, window function, and percentile method documented and version-controlled. (2) Frequency-response flatness validated by Tier 3 calibration across the full analysis span, with residual ripple < 0.5 dB. (3) RBW equivalent characterised and recorded in measurement metadata. (4) Interference control within analysis span verified against a defined co-channel rejection criterion. (5) I/Q imbalance correction residual verified < -40 dBc at the target channel frequency; results exceeding this threshold shall trigger <code>IQ_RESIDUAL</code> and invalidate bandwidth estimates. ^c (6) Cross-validation against reference instrumentation (minimum one traceable signal generator sweep) completed and archived prior to first compliance-use deployment.
Adjacent-Channel / Out-of-Band Emission	Unsupported	Disqualifying limitations: SFDR of 50–55 dBc is insufficient for regulatory emission masks requiring > 70 dB attenuation; the gap cannot be closed by calibration. Reclassification requires an external preselector or SAW filter <i>and</i> a 14–16 bit ADC front-end; both are required, not optional.
Peak Deviation / Multiplex Power	Screening	FM demodulator linearity and deviation calibration required. Multiplex-power estimation valid only when baseband conditioning (de-emphasis, pilot extraction) is validated against a reference modulation analyser. Pilot and RDS sub-carrier magnitudes shall be verified independently.
Channel Occupancy	Conditional	Mandatory preconditions — all must be satisfied and at-

Reference Requirements for Full Compliance-Grade Deployments

The following requirements apply to each measurand when compliance-grade or Conditional Compliance-Grade reporting is sought. Requirements for measurands classified as Unsupported on the baseline HackRF One platform (field strength, ACLR) are scoped to upgraded or augmented platforms as defined in the Upgrade Paths section; they do not imply that these measurands are achievable on the unaugmented HackRF One. All requirements are in addition to, not in substitution for, the preconditions listed in Table 3.

- *Carrier Frequency Error and Stability.* • *Reference clock traceability.* The SDR master clock shall be disciplined by a GPS-Disciplined Oscillator (GPSDO) or equivalent traceable reference providing a fractional frequency accuracy of $\pm 1 \times 10^{-9}$ or better. An external Rubidium oscillator is acceptable only when it is itself GPS-steered or carries a current calibration certificate traceable to a national frequency standard; a free-running Rubidium standard does not satisfy this requirement.
- *Estimation algorithm.* Carrier centre frequency shall be estimated using the Stage 5 pipeline specified in the Estimation Pipeline section (parabolic interpolation primary, power centroid and edge midpoint as supporting estimators). The power-weighted centroid may be used as the primary estimator only when the peak-parabolic method is flagged invalid by the Stage 5 validity logic.
- *Guard-band decision logic.* The Guard-Band rule (Eq. 4) shall be applied using the 2 kHz carrier-frequency tolerance specified by FCC Part 73 or the applicable jurisdictional limit. If $|f_{\text{cal}} - f_{\text{target}}| + U \leq T$, the result is compliant; if $|f_{\text{cal}} - f_{\text{target}}| - U > T$, non-compliant; otherwise uncertain and subject to the escalation procedure in Table 4.
- *Uncertainty-driven escalation.* As specified in Table 4, results in the uncertain zone shall trigger a repeated measurement cycle with $4\times$ extended averaging. If the result remains uncertain after three attempts, it shall be logged as UNRESOLVED and escalated to operator review.
- *Reference drift monitoring.* The GPSDO lock state and 1 PPS timing validity shall be recorded continuously. Any measurement acquired during a GPSDO unlock event shall be flagged UNCALIBRATED_REFERENCE and classified as screening-grade. Temperature of the SDR front-end shall be logged as a diagnostic field; if the node operates without a disciplined reference (e.g., during holdover), the estimated frequency drift based on the TCXO drift profile shall be recorded as an additional Type B uncertainty contribution.
- *Calibrated Received Power.* • *Reference plane traceability.* The receive chain shall undergo Tier 1 laboratory calibration establishing absolute gain accuracy at the defined reference plane (antenna port or SDR input bulkhead). Tier 2 field verification shall confirm that the node has not drifted beyond acceptable limits relative to the Tier 1 state; Tier 2 alone does not establish absolute traceability.
- *Shared-risk decision logic.* Expanded uncertainty U ($k = 2$) shall be computed in real time. Values falling within the U -interval of the applicable ERP or received-power limit shall be flagged for verification using a calibrated mobile unit or a compliance-grade co-located node, per Table 4.
- *Integration method.* Calibrated received power shall be computed as the RMS-weighted PSD integral over the defined analysis bandwidth B (nominally 200 kHz for FM broadcast), after gain-state and frequency-response correction.

- *Field Strength (upgraded or augmented platforms only).* • *Scope.* Field strength is classified Unsupported on the baseline HackRF One. The following requirements apply only to platforms satisfying the reclassification criteria stated in Table 3 (Field Strength row).
 - *Antenna factor application.* Frequency-dependent antenna factors (AF) shall be applied to convert calibrated terminal power to field strength in dB μ V/m. The AF table shall be derived from a Tier 1 antenna calibration and shall cover the full operating bandwidth at the deployment site.
 - *Reference plane.* The field-strength reference plane is the antenna terminals. All feedline losses and connector corrections shall be characterised and applied as part of the Tier 1 calibration.
- *Occupied Bandwidth.* • *Windowing.* Blackman-Harris windowing shall be used for all occupied bandwidth measurements, consistent with the Stage 3 spectral estimation pipeline. Flat-top windowing shall not be used; its non-zero passband ripple introduces systematic bias in the power integration used for the $\beta\%$ boundary determination.
 - *RBW correction.* Normalised compensation for the equivalent noise bandwidth (ENBW) of the analysis window shall be applied before the $\beta\%$ power boundary is computed.
 - *Adaptive resolution.* If the $\beta\%$ (99%) boundary falls within 5 kHz of the regulatory mask limit, the system shall increase the FFT size to achieve finer bin resolution. Any change to the FFT size shall be logged in the measurement metadata and treated as a processing configuration change subject to version control, consistent with the Conditional Compliance precondition (1) in Table 3.
 - *I/Q residual gate.* Before reporting an occupied bandwidth result, the system shall verify that the IQ_RESIDUAL flag is not set. Results acquired when the corrected mirror power exceeds –40 dBc shall be invalidated per Table 3 precondition (5).
- *Adjacent-Channel and Out-of-Band Emission (upgraded platforms only).* • *Scope.* ACLR is classified Unsupported on the baseline HackRF One. The following requirements apply only to platforms satisfying the reclassification criteria stated in Table 3 (ACLR row).
 - *SFDR margin gate.* Before any ACLR measurement is reported, the system shall verify that the platform’s in-situ SFDR exceeds the applicable regulatory mask limit by at least 15 dB. If this condition is not met, the measurement shall be suppressed and flagged UNSUPPORTED.
 - *Integration bandwidth.* The integration window shall be defined as 200 kHz centred on the adjacent or alternate channel carrier, consistent with the FM channel plan.
 - *Uncertain results.* Uncertain ACLR results shall not support a non-compliance determination. Escalation shall follow the array-level coordination procedure in Section 5 using a higher-dynamic-range validation node where available.
- *Peak Deviation and Multiplex Power.* • *Demodulation bandwidth.* FM demodulation shall use a minimum IF bandwidth of 400 kHz to capture high-order sidebands. This requirement is consistent with the measurement method defined in ITU-R BS.412 for the applicable deviation and multiplex power indicators.
 - *MPX power reference.* Integration of the multiplex signal power shall be reported in dBr relative to the 0 dBr reference level, defined as the power of the composite MPX signal

at 75 kHz peak deviation of a 1 kHz sine tone, with the 19 kHz stereo pilot present at its nominal level. This definition shall be recorded in the measurement procedure.

- *Sub-carrier verification.* The 19 kHz pilot tone and 57 kHz RDS sub-carrier shall be independently verified in magnitude. Results shall be reported relative to the main carrier level and stored as supporting evidence.
- *Channel Occupancy.*
 - *Detection threshold.* The occupancy threshold shall be a site-calibrated offset above the locally verified noise floor, derived from the Tier 2 noise-floor verification required by Table 3 precondition (2). A fixed default offset shall not be used for Conditional Compliance-Grade reporting.
 - *Statistical reporting.* Occupancy shall be reported as the fraction of observation windows in which the channel is declared occupied under the documented detection rule, together with a confidence interval derived from the characterised false-alarm and missed-detection probabilities. The reporting model shall be consistent with the statistical probability framework required by Table 3 precondition (3).
 - *Hysteresis and dwell.* Temporal hysteresis shall be applied to prevent false occupancy triggers from intermittent burst signals. The hysteresis parameters shall be fixed in the version-controlled processing configuration per Table 3 precondition (4).

Estimation Pipeline of Measurands

Reference Carrier-Frequency Estimation Procedure. The processing pipeline is specified for estimating the carrier centre frequency of an FM broadcast signal from complex baseband samples acquired with a HackRF One software-defined radio receiver. The implementation is organised as six sequential stages. The pipeline is explicitly designed to retain intermediate quantities so that spectral estimates, candidate selection, carrier estimates, and confidence scores remain auditable and can be reinterpreted if calibration metadata are updated.

- *Stage 1: Acquisition Configuration.* The acquisition stage constructs a validated hardware configuration for one HackRF capture session. In the notebook implementation, the receiver is configured with

$$f_{\text{LO}} = f_{\text{target}} + f_{\text{off}}, \quad f_{\text{off}} = \frac{f_s}{4},$$

so the observed FM channel is offset from the DC bin by one quarter of the sample rate. The configuration validator additionally requires

$$|f_{\text{off}}| \geq 300 \text{ kHz}, \quad |f_{\text{off}}| + 150 \text{ kHz} < \frac{f_s}{2},$$

so that the target FM allocation remains well separated from the DC region and inside the usable complex baseband span.

The user-facing HackRF receive controls are:

- RF amplifier state `amp_enable` (internal +14 dB front-end amplifier),
- IF gain `if_gain_db` in 8 dB steps over 0 to 40 dB,
- baseband gain `baseband_gain_db` in 2 dB steps over 0 to 62 dB.

For compatibility with older naming, legacy fields `lna_gain_db` and `vga_gain_db` are accepted and mapped to IF gain and baseband gain respectively. The configuration records

$$f_{\text{target}}, \quad f_{\text{off}}, \quad f_{\text{LO}}, \quad f_s,$$

together with the individual gain settings and the total receiver gain

$$G_{\text{RX}} = G_{\text{IF}} + G_{\text{BB}} + \begin{cases} 14 \text{ dB}, & \text{if RF amp is enabled,} \\ 0, & \text{otherwise.} \end{cases}$$

A warning is emitted when the total gain exceeds the notebook warning level of 32 dB, but clipping prevention is not enforced analytically at this stage.

During capture, the HackRF is configured in the sequence sample-rate $\rightarrow$ frequency $\rightarrow$ gain chain. A short tuning settling pause is inserted after setting the LO. Raw interleaved `int8` I/Q data are collected, converted to complex baseband as

$$x[n] = \frac{I[n] + jQ[n]}{128},$$

and a warm-up segment of 131,072 complex samples is discarded before the final retained block is formed. Clipping is detected directly from the raw ADC stream by checking whether any sample equals 127 or -128 , and this boolean clipping outcome is propagated to later stages.

- *Stage 2: Preprocessing.* The preprocessing stage operates on the retained complex baseband sequence and performs three sub-stages in fixed order: DC-spike mitigation, I/Q imbalance correction, and relative noise-floor estimation.

- 2a. *DC-spike mitigation.* The notebook removes residual DC contamination by processing the signal in non-overlapping FFT blocks of length $N_{\text{DC}} = 4096$ by default. For each complete block, the spectrum is FFT-shifted, a symmetric guard region centred on the DC bin is identified, and the guarded samples are replaced by interpolated amplitude and phase derived from the adjacent non-guard context bins. With guard width $N_{\text{guard}} = 5$ and interpolation context $N_{\text{ctx}} = 4$, the guarded frequency span recorded in metadata is

$$W_{\text{DC}} = \frac{(2N_{\text{guard}} + 1)f_s}{N_{\text{DC}}}.$$

The implementation stores the mean pre-correction DC-bin power across the processed blocks as a diagnostic field. Any trailing remainder shorter than one FFT block is passed through unchanged.

- 2b. *I/Q imbalance correction.* The corrected complex sequence is then passed through a 2×2 widely-linear correction matrix derived from the current `IQCalibration` object:

$$\tilde{\mathbf{x}}[n] = \mathbf{C} \mathbf{x}[n], \quad \mathbf{C} = \begin{bmatrix} 1 & 0 \\ -\sin \phi & \cos \phi / g \end{bmatrix},$$

where g is the amplitude-imbalance factor and ϕ is the phase error in radians. The notebook also provides an `estimate_iq_calibration()` helper that estimates g and ϕ from a reference tone, but the correction stage itself simply applies the supplied calibration.

Mirror-rejection diagnostics are computed only when an explicit evaluation tone offset is supplied. In that case the notebook measures mirror power before and after correction, reports the suppression in dB, and raises an I/Q residual flag when the corrected mirror remains above the notebook threshold of -40 dBc. When no diagnostic tone is supplied, mirror-power metrics are stored as NaN and no residual flag is asserted.

- 2c. *Relative noise-floor estimation.* The final preprocessing sub-stage estimates a *relative*, pre-calibration noise baseline. The corrected sequence is partitioned into up to $K = 16$ complete, non-overlapping frames of length $N_{\text{noise}} = 4096$ by default. Each frame is windowed by a Blackman–Harris window normalized to unit RMS and transformed to the frequency domain. The frame power spectra $\{P_m(k)\}$ are accumulated, and the provisional baseline is formed from the per-bin percentile, with default percentile $p = 10$:

$$N_{\text{prov}}(k) = \text{percentile}_{10}\{P_m(k)\}_{m=1}^K.$$

The notebook then protects the baseline against signal occupancy. It forms the mean spectrum across the same frames, converts both the provisional percentile baseline and the mean spectrum to dB, and marks as occupied any bins where the mean exceeds the provisional baseline by more than 6 dB. That occupied mask is dilated by 12 bins on each side, the masked baseline is linearly interpolated across the remaining unmasked bins, and the result is smoothed by a reflected moving-average filter of width 33 bins. If the protected mask would leave fewer than 10% of bins unmasked, the notebook falls back to a smoothed version of the provisional baseline instead of interpolating through the mask.

The Stage 2 output therefore contains:

- the DC-corrected and I/Q-corrected sample stream,
- the inclusive DC guard-band width,
- I/Q imbalance diagnostics,
- a *relative* frequency-dependent noise-floor estimate `noise_floor_relative_db`,

not yet a calibrated dBm/Hz baseline.

- *Stage 3: Spectral Estimation.* Stage 3 computes the calibrated wideband power spectral density used by the downstream detector and carrier estimator. The implementation requires

$$N_{\text{FFT}} \geq 8192, \quad N_{\text{FFT}} \text{ power-of-two}, \quad M \geq 4$$

complete non-overlapping averaging frames. The default configuration is $N_{\text{FFT}} = 8192$, $M = 16$, and a Blackman–Harris analysis window. A boxcar window is explicitly rejected because the notebook treats its sidelobe suppression as inadequate for FM monitoring.

For each complete frame m , the PSD estimate is computed as

$$\hat{P}_m(k) = \frac{\left| \sum_{n=0}^{N_{\text{FFT}}-1} x_m[n] w[n] e^{-j2\pi kn/N_{\text{FFT}}} \right|^2}{f_s \sum_{n=0}^{N_{\text{FFT}}-1} |w[n]|^2},$$

and Welch averaging yields

$$\hat{P}(k) = \frac{1}{M} \sum_{m=1}^M \hat{P}_m(k).$$

The resulting average is first represented in dBFS/Hz, then converted to dBm/Hz by the scalar gain-calibration term

$$C_{\text{gain}} = (-10 \text{ dBm}) - G_{\text{RX}} + L_{\text{cable}} - G_{\text{ant}},$$

with the antenna term applied only when antenna correction is explicitly enabled. An optional per-bin frequency-response correction vector may also be applied when its length matches the FFT size.

The Stage 3 frequency axis is constructed as

$$f_k = f_{\text{baseband},k} + f_{\text{LO}} - f_{\text{off}},$$

so the Stage 1 tuning offset is already removed at this point. This is the frequency axis used by both Stage 4 and Stage 5.

The Stage 2 relative noise floor is then resampled from the Stage 2 FFT grid to the Stage 3 FFT grid by frequency interpolation. The calibrated Stage 3 noise floor is formed by adding both the Stage 3 scalar gain correction and the Stage 2 density offset

$$-10 \log_{10}(f_s N_{\text{noise}}),$$

yielding the stored field `noise_floor_calibrated_dbm_hz`. The notebook also retains the resampled relative baseline separately as `noise_floor_relative_db`. Shape and finiteness checks are applied to both arrays before Stage 3 returns.

- *Stage 4: Channel Candidate Detection.* Stage 4 does not search for isolated spectral lines. Instead, it detects occupied above-threshold regions in the calibrated PSD. A smoothed baseline threshold is formed from the calibrated noise floor:

$$T(k) = \tilde{N}_{\text{cal}}(k) + \Delta_{\text{det}},$$

with default detection margin

$$\Delta_{\text{det}} = 8 \text{ dB},$$

and allowed bounds 6 to 20 dB. The threshold itself is smoothed by a reflected moving-average filter of width 9 bins.

The occupied-mask rule is

$$\hat{P}_{\text{cal}}(k) > T(k).$$

Short gaps in that mask are closed when their width does not exceed the equivalent of 15 kHz. Regions shorter than the equivalent of 40 kHz are discarded.

For each surviving region $[k_s, k_e]$, the notebook computes:

- peak bin and peak power inside the region;
- median calibrated noise floor over the region;
- region SNR as peak power minus regional median noise floor;
- raw occupied bandwidth

$$W_{\text{occ}} = f[k_e] - f[k_s];$$

- edge midpoint

$$f_{\text{edge}} = \frac{1}{2}(f[k_s] + f[k_e]);$$

- power centroid obtained from the positive linear-power excess above the calibrated noise floor across the region;
- shoulder power, defined as the maximum PSD in the left or right outer shoulder windows extending up to ± 400 kHz beyond the region;
- region prominence

$$\Pi = P_{\text{peak}} - P_{\text{shoulder}}.$$

The Stage 4 candidate centre frequency stored in the candidate object is *the edge midpoint*, not the peak-bin frequency and not the centroid. A bandwidth-out-of-range flag is raised when

$$W_{\text{occ}} \notin [40 \text{ kHz}, 400 \text{ kHz}].$$

A candidate is considered a valid carrier only when it is not marked as an artifact, is not in DC proximity, is not width-rejected, and has

$$\Pi \geq 10 \text{ dB}.$$

The notebook includes three artefact checks:

- (i) *DC proximity*: in the currently executed `run_stage4(...)` code path, the default argument `dc_guard_band_hz` is 30 kHz, and the implemented flag test is

$$|f - f_{\text{target}}| < 2 \times 30 \text{ kHz}.$$

Hence, in the executed integration path, DC proximity is effectively tested within 60 kHz of the target frequency, not from a Stage 2-derived W_{DC} expression.

- (ii) *Mirror-image test*: in the current executed `run_stage4(...)` code path, candidate pairs are compared around the reflection axis

$$f_{\text{refl}} = f_{\text{target}}.$$

A candidate is marked as a mirror only when its paired reflected candidate lies within 5 kHz of the expected mirror location, differs in peak power by at least 6 dB, and has sufficiently similar bandwidth (minimum width ratio 0.5).

- (iii) *Harmonic test*: harmonic flagging is present only as a placeholder and currently performs no rejection.

Each candidate is ranked by the notebook score

$$S = 0.45 S_{\text{prox}} + 0.25 S_{\text{snr}} + 0.20 S_{\text{width}} + 0.10 S_{\text{prom}},$$

where

$$S_{\text{prox}} = \exp\left(-\frac{|f_c - f_{\text{target}}|}{150 \text{ kHz}}\right),$$

$$S_{\text{snr}} = \text{clip}\left(\frac{\text{SNR}}{25}, 0, 1\right), \quad S_{\text{width}} = \begin{cases} 1, & W_{\text{occ}} \in [100, 250] \text{ kHz}, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$S_{\text{prom}} = \text{clip}\left(\frac{\Pi}{20 \text{ dB}}, 0, 1\right).$$

Candidates closer than 100 kHz are deduplicated by keeping the higher-scored candidate. The return list is ordered as all valid carriers in descending score, followed by invalid candidates in descending score, and is truncated to at most 8 entries.

- *Stage 5: Carrier-Frequency Estimation.* Stage 5 estimates the carrier frequency from the primary valid candidate returned by Stage 4 unless a different candidate is explicitly supplied.

- *Primary estimator: peak-centric parabolic interpolation.*

The primary estimator is a three-point parabolic interpolation around the peak bin inside the selected occupied region. The notebook accepts the Stage 4 threshold array only for shape validation and API compatibility; it is *not* used as a Stage 5 peak-validity gate. The validity logic is purely geometric and local.

Let k^* be the maximum-PSD bin inside the region. The estimator first requires that k^* is not on the global spectral boundary and that the peak is a local maximum in the dB domain:

$$y_0 \geq y_{-1}, \quad y_0 \geq y_{+1},$$

with local prominence

$$y_0 - \max(y_{-1}, y_{+1}) \geq 0.5 \text{ dB}.$$

The quadratic refinement is then carried out in *linear power*:

$$p_{-1} = 10^{y_{-1}/10}, \quad p_0 = 10^{y_0/10}, \quad p_{+1} = 10^{y_{+1}/10},$$

$$d = p_{-1} - 2p_0 + p_{+1}, \quad \delta = \frac{1}{2} \frac{p_{-1} - p_{+1}}{d}.$$

The curvature must be finite, non-degenerate, and concave ($d < 0$). The raw sub-bin offset must satisfy

$$|\delta| \leq 1.$$

If so, it is clipped conservatively into

$$\delta \in [-0.5, +0.5]$$

before the refined frequency is reported:

$$\hat{f}_{\text{peak}} = f[k^*] + \delta \Delta f_{\text{bin}}.$$

If any of the above checks fails, the peak-parabolic method is marked invalid and Stage 5 falls back to a supporting estimator.

- *Supporting estimators.*

Two supporting estimators are always computed on the calibrated spectral arrays:

- *Edge midpoint:* Stage 5 recomputes the left and right threshold crossings using the calibrated PSD and the calibrated Stage 4 threshold, linearly interpolates both crossings, and reports the midpoint frequency and interpolated edge bandwidth.
- *Power centroid:* Stage 5 computes a power centroid using the positive linear-power excess above the *calibrated* noise floor within the selected region.

- *Optional phase-domain refinement.*

The notebook preserves API switches for MUSIC and Pisarenko refinement but intentionally leaves phase-domain refinement disabled in the compliance-tuned Stage 5 path. The corresponding return field is therefore **None**.

– *WBFM-aware repeatability diagnostic.*

Legacy per-frame instantaneous frequency statistics are intentionally disabled in the current notebook because they are not physically meaningful for WBFM carrier-centre estimation. The fields `frame_frequency_estimates_hz`, `frame_std_hz`, `frame_mean_hz`, and `frame_inconsistency_flag` are retained for compatibility but are set to

$$[], \quad 0, \quad \hat{f}_{\text{primary_fallback}}, \quad \text{False},$$

respectively, where $\hat{f}_{\text{primary_fallback}}$ equals the peak-parabolic estimate when that estimate is valid, and otherwise equals the centroid estimate.

Instead, repeatability is measured across repeated *averaged* PSD estimates built from non-overlapping blocks of the Stage 3 frame spectra. The block count defaults to 4, limited by however many frame spectra are available. Within each block, the stored per-frame PSDs are converted from dB to linear power, averaged, converted back to dB, and re-estimated by the same Stage 5 method cascade: peak-parabolic $\rightarrow$ centroid $\rightarrow$ edge midpoint $\rightarrow$ fallback frequency. The resulting list $\{\hat{f}_{\text{block},i}\}$ yields

$$\sigma_{\text{block}} = \text{std}\{\hat{f}_{\text{block},i}\}, \quad \mu_{\text{block}} = \text{mean}\{\hat{f}_{\text{block},i}\}.$$

A block-repeatability warning flag is raised when

$$\sigma_{\text{block}} > 1000 \text{ Hz}.$$

To ensure consistency with the stored Stage 3 frame spectra, the block repeatability estimator converts the calibrated Stage 3 noise floor and threshold back to the frame-relative domain by subtracting the Stage 3 scalar gain correction before applying the repeated-estimation cascade.

– *Final estimator selection.*

The notebook selects the final raw estimate as follows:

- (i) use the peak-parabolic estimate if valid;
- (ii) otherwise use the centroid if finite;
- (iii) otherwise use the edge midpoint.

A fallback flag and a fallback-reason string are emitted whenever the peak-parabolic estimate is invalid.

– *Frequency correction and deviation.*

The final calibrated estimate is

$$\hat{f}_{\text{cal}} = \hat{f}_{\text{raw}} + \varepsilon_{\text{cal}},$$

where ε_{cal} is the residual frequency-calibration correction supplied through `FrequencyCalibration`. No additional offset-tuning subtraction is applied in Stage 5 because the Stage 3 frequency axis has already been corrected for f_{off} .

The reported carrier deviation from the assigned frequency is

$$\Delta f = \hat{f}_{\text{cal}} - f_{\text{target}}.$$

– *Uncertainty estimate.*

The Stage 5 residual-frequency uncertainty is a conservative maximum of:

- a minimum interpolation residual equal to 5% of the FFT bin spacing,
- the supplied calibration uncertainty,
- the block-repeatability standard deviation,
- pairwise disagreement terms between the available Stage 5 methods.

Formally, the notebook returns

$$u_f = \max\left(0.05 \Delta f_{\text{bin}}, u_{\text{cal}}, \sigma_{\text{block}}, |\hat{f}_{\text{peak}} - \hat{f}_{\text{cent}}|, |\hat{f}_{\text{peak}} - \hat{f}_{\text{edge}}|, |\hat{f}_{\text{edge}} - \hat{f}_{\text{cent}}|\right),$$

with the peak-related terms included only when the peak-parabolic estimate is valid.

- *Stage 6 — Confidence Scoring and Result Assembly.* The notebook computes a composite confidence scalar

$$C \in [0, 1]$$

from six normalized sub-scores:

$$C = 0.25 c_{\text{agree}} + 0.25 c_{\text{block}} + 0.15 c_{\text{geom}} + 0.15 c_{\text{region}} + 0.10 c_{\text{margin}} + 0.10 c_{\text{cal}}.$$

The six sub-scores are defined exactly as follows.

- *Agreement score* c_{agree} .

The notebook computes the method spread as the range of the finite values among

$$\hat{f}_{\text{peak}}, \quad \hat{f}_{\text{cent}}, \quad \hat{f}_{\text{edge}},$$

and then sets

$$c_{\text{agree}} = \exp\left(-\left(\frac{\Delta f_{\text{spread}}}{1500 \text{ Hz}}\right)^2\right).$$

- *Block-repeatability score* c_{block} .

Using the Stage 5 block-repeatability standard deviation,

$$c_{\text{block}} = \exp\left(-\left(\frac{\sigma_{\text{block}}}{500 \text{ Hz}}\right)^2\right).$$

- *Peak-geometry score* c_{geom} .

This score combines the local parabolic-peak prominence and the absolute sub-bin offset:

$$c_{\text{geom}} = \text{clip}\left(\frac{\Pi_{\text{local}}}{2 \text{ dB}}, 0, 1\right) \cdot \exp\left(-\left(\frac{|\delta|}{0.7}\right)^4\right).$$

- *Region-definition score* c_{region} .

The score starts at 1 and is multiplied by 0.7 for each missing edge, multiplied by another 0.7 if the occupied bandwidth lies outside the notebook ideal interval

$$[80 \text{ kHz}, 250 \text{ kHz}],$$

and finally multiplied by the asymmetry penalty

$$\exp\left(-\left(\frac{|\hat{f}_{\text{edge}} - \hat{f}_{\text{cent}}|}{5000 \text{ Hz}}\right)^2\right).$$

– *Baseline-margin score* c_{margin} .

The Stage 5 peak-above-baseline margin is mapped as

$$c_{\text{margin}} = \text{clip}\left(\frac{\Pi_{\text{baseline}} - 6 \text{ dB}}{10 \text{ dB}}, 0, 1\right).$$

– *Calibration score* c_{cal} .

The calibration-quality term is

$$c_{\text{cal}} = \exp\left(-\left(\frac{u_{\text{cal}}}{1000 \text{ Hz}}\right)^2\right).$$

The Stage 6 quality-flag word is assembled from the carrier estimate, the preprocessing result, and the acquisition result. The notebook may set:

- DC_PROXIMITY,
- IQ_RESIDUAL,
- LOW_SNR when SNR < 10 dB,
- GAIN_SATURATION when raw-sample clipping was detected,
- BLOCK_REPEATABILITY,
- METHOD_DISAGREEMENT,
- FALLBACK_ESTIMATOR_USED,
- UNCALIBRATED_REFERENCE.

A reference is treated as uncalibrated when the reference-source field is empty or one of {uncalibrated, unknown, none}, or when the calibration-state identifier is empty or one of {default, unknown, none}.

The final Stage 6 result packages:

- the calibrated carrier estimate and its deviation from target,
- SNR, occupied bandwidth, edge bandwidth, and noise-floor-at-peak,
- the full confidence breakdown,
- DC and I/Q diagnostics,
- Stage 5 estimator details and disagreement fields,
- FFT and averaging metadata,
- calibration metadata and reference source,
- an observation-window duration.

When Stage 6 is supplied with explicit Stage 3 FFT size and averaging-frame count, the observation window is computed as

$$T_{\text{obs}} = \frac{N_{\text{FFT}} M}{f_s};$$

otherwise it defaults to the retained acquisition length divided by the sample rate.

Uncertainty Budget and Reporting

Uncertainty Classification. Measurement uncertainty shall be evaluated and reported in accordance with the *Guide to the Expression of Uncertainty in Measurement* (GUM, JCGM 100:2008). Every reported compliance measurand shall be accompanied by a documented, itemised uncertainty budget that identifies all contributing sources, their type, their standard uncertainty value, and the calibration tier from which they are derived. The budget shall be archived with the measurement record and shall be traceable to the calibration certificate of the node that produced the result.

Uncertainty contributions are classified as follows.

- *Type A uncertainty* (u_A). Evaluated by statistical analysis of repeated observations under repeatability conditions. Type A contributions include thermal noise, digitisation jitter, environmental fluctuations, and algorithmic repeatability. The standard uncertainty is the experimental standard deviation of the mean of n repeated observations:

$$u_A = \frac{s}{\sqrt{n}}, \quad s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}. \quad (1)$$

The number of observations n shall be sufficient to make the Type A contribution statistically meaningful; a minimum of $n = 10$ independent estimates is recommended for compliance-grade measurands.

- *Type B uncertainty* (u_B). Evaluated by means other than statistical analysis, including calibration certificates, manufacturer specifications, prior measurement data, engineering judgement, and physical modelling. Type B contributions include calibration residuals, reference-standard uncertainty, gain non-linearity, frequency-response flatness, and temperature drift. The standard uncertainty for a Type B contributor with stated limits $\pm a$ and assumed distribution p is:

$$u_B = \begin{cases} a/\sqrt{3}, & \text{rectangular distribution,} \\ a/\sqrt{6}, & \text{triangular distribution,} \\ a/2, & \text{normal distribution at } k = 2. \end{cases} \quad (2)$$

The assumed distribution shall be stated and justified in the uncertainty budget for each contributor.

Combined and Expanded Uncertainty. The combined standard uncertainty u_c is obtained by combining all Type A and Type B standard uncertainty components. When all contributions are mutually independent, the combination reduces to a root-sum-square:

$$u_c = \sqrt{\sum_i u_{A,i}^2 + \sum_j u_{B,j}^2}. \quad (3)$$

When two or more Type B contributors share a common physical cause — for example, gain non-linearity and frequency-response flatness both arising from the same front-end hardware — the independence assumption is not valid. In such cases the full GUM covariance form shall be used:

$$u_c^2 = \sum_i u_i^2 + 2 \sum_i \sum_{j>i} r(u_i, u_j) u_i u_j, \quad (4)$$

where $r(u_i, u_j)$ is the estimated correlation coefficient between contributions u_i and u_j . The presence and magnitude of significant correlations shall be assessed and documented in the uncertainty budget. If correlations are unknown, they shall be assumed to be zero with a documented justification; this assumption shall be flagged as a limitation in the associated compliance record.

The expanded uncertainty U is obtained by multiplying the combined standard uncertainty by a coverage factor k :

$$U = k \cdot u_c. \quad (5)$$

For regulatory reporting, $k = 2$ shall be used as the default, corresponding to an approximately 95% confidence interval under a normal distribution. When the uncertainty budget is dominated by Type B contributors with non-Gaussian distributions (e.g., rectangular or triangular), the effective degrees of freedom ν_{eff} shall be estimated using the Welch-Satterthwaite formula and k selected from the t -distribution at the 95.45% level for ν_{eff} degrees of freedom. The value of k actually used shall be recorded in every compliance report alongside U .

Periodic Re-Evaluation. The uncertainty budget for each deployed node and each reported measurand shall be re-evaluated whenever any of the following conditions occurs:

- A Tier 1 or Tier 2 calibration is performed or a calibration certificate is renewed;
- A hardware change is made to any component of the receive chain, including antenna, feedline, LNA, filter, or SDR unit;
- A software or firmware update affects any stage of the processing pipeline that contributes to the measurand estimate;
- Tier 2 field verification reveals a drift exceeding the defined warning limit;
- Environmental conditions at the deployment site change in a way that may affect any Type B contributor (e.g., seasonal temperature range, installation of nearby transmitters).

Re-evaluation results shall be archived with a timestamp, the triggering condition, and the revised uncertainty budget. If the revised expanded uncertainty U exceeds the guard-band width for any compliance-grade measurand, the affected measurand shall be downgraded to Screening-Grade until the uncertainty is reduced to an acceptable level or the decision rule is revised under documented authority.

Uncertainty-Aware Compliance Decisions. When comparing a calibrated measurand estimate x against a regulatory threshold T at reference point x_{ref} , the system shall apply one of the three decision rules defined formally in Equations 6–???. This paragraph provides a normative summary; the formal mathematical definitions, boundary conditions, and applicability constraints in the following section are authoritative and take precedence in case of any ambiguity.

- *Simple threshold rule (non-conservative).* $|x - x_{\text{ref}}| \leq T \Rightarrow$ compliant; $|x - x_{\text{ref}}| > T \Rightarrow$ non-compliant. This rule does not account for measurement uncertainty and shall be used only when the expanded uncertainty satisfies $U < 0.1 T$, or when the applicable regulatory framework explicitly requires an uncertainty-blind comparison. It shall not be applied to any measurand for which $U \geq 0.1 T$ without documented justification and explicit authority from the responsible regulatory body.

- *Guard-band rule (conservative, ISO 14253-1).* $|x - x_{\text{ref}}| + U \leq T \Rightarrow \text{compliant}$; $|x - x_{\text{ref}}| - U > T \Rightarrow \text{non-compliant}$; otherwise $\Rightarrow \text{uncertain}$. Results in the uncertain zone $[T - U, T + U]$ shall be escalated per the measurand-specific procedure in Table ???. This rule minimises the risk of false compliance and is the default for carrier-frequency error, occupied bandwidth, and adjacent-channel emission.
- *Shared-risk rule (equal false-accept and false-reject risk).* Acceptance zone: $x_{\text{ref}} \pm (T - U)$; rejection zone: $|x - x_{\text{ref}}| > T + U$; uncertain zone: $T - U < |x - x_{\text{ref}}| \leq T + U$. Results in the uncertain zone shall be flagged for verification using a calibrated mobile unit or a compliance-grade validation node; they shall not be resolved by operator qualification alone without corroborating evidence. This rule is the default for calibrated received power.

The adopted decision rule shall be documented in the applicable compliance procedure, recorded in the metadata of every compliance determination, and shall not be changed between measurement cycles for the same station and measurand without a documented configuration change record. The measurand-specific default rule and the handling procedure for uncertain outcomes are specified in Table ???.

Compliance Decision Logic and Uncertainty Handling

Three decision frameworks are supported. The applicable rule for each primary measurand is specified in Table 4. The rule shall be selected before measurement begins, documented in the compliance procedure, and shall not be changed between measurement cycles for the same station and measurand pair without a configuration change record.

- *Simple Threshold Rule (zero guard band).*

$$\text{Decision} = \begin{cases} \text{COMPLIANT}, & |x - x_{\text{ref}}| \leq T, \\ \text{NON-COMPLIANT}, & |x - x_{\text{ref}}| > T, \end{cases} \quad (6)$$

where x_{ref} is the regulatory reference point (e.g., assigned carrier frequency, 0 dBm reference level) and T is the applicable regulatory threshold. This rule does **not** account for measurement uncertainty and shall be used only when:

- $U < 0.1 T$, so that uncertainty is negligible relative to the threshold, or
- the regulatory framework explicitly requires an uncertainty-blind comparison.

It shall not be applied when $U \geq 0.1 T$ without documented justification and explicit authority from the responsible regulatory body.

- *Conservative Guard-Band Rule (ISO 14253-1).*

$$\text{Decision} = \begin{cases} \text{COMPLIANT}, & |x - x_{\text{ref}}| + U \leq T, \\ \text{NON-COMPLIANT}, & |x - x_{\text{ref}}| - U > T, \\ \text{UNCERTAIN}, & \text{otherwise.} \end{cases} \quad (7)$$

This rule minimises the risk of false compliance by requiring the worst-case bound $|x - x_{\text{ref}}| + U$ to remain within the limit. Results falling in the uncertain zone $[T - U, T + U]$ shall be handled per Table 4.

– *Shared-Risk Rule (equal false-accept and false-reject risk).* Define:

$$\begin{aligned}
A &= [x_{\text{ref}} - (T - U), x_{\text{ref}} + (T - U)], \\
R &= (-\infty, x_{\text{ref}} - (T + U)) \cup (x_{\text{ref}} + (T + U), \infty). \\
\text{Decision} &= \begin{cases} \text{COMPLIANT}, & x \in A, \\ \text{NON-COMPLIANT}, & x \in R, \\ \text{UNCERTAIN}, & x \notin A \cup R. \end{cases} \tag{8}
\end{aligned}$$

The uncertain zone is the symmetric annular region $(T - U) < |x - x_{\text{ref}}| \leq (T + U)$; points on the boundary $|x - x_{\text{ref}}| = T + U$ are assigned to **NON-COMPLIANT**. This rule is appropriate when the economic or operational consequences of false acceptance and false rejection are comparable.

Measurand-Specific Decision Rules. Table 4 specifies the default decision rule and the handling procedure for uncertain outcomes for each primary compliance measurand. The table is normative; deviations require documented authority.

Policy Override and Escalation. When a measurand result remains in the uncertain zone after all handling procedures in Table 4 have been exhausted, the system shall execute the following escalation sequence:

1. Record the result with **UNRESOLVED** status, including the measurand value, expanded uncertainty U , decision rule applied, number of repeated attempts, and the reason the uncertain zone could not be resolved.
2. Generate an escalation alert containing all supporting evidence: the bounded raw I/Q excerpt associated with the triggering acquisition window, the calibrated PSD estimate, the itemised uncertainty budget, and the current calibration state of the node.
3. Suspend automated compliance determination for the affected station and measurand pair. The suspension shall be time-bounded: if operator review or validation instrumentation deployment has not resolved the result within the interval defined in the system operating procedure (default: 72 hours), the result shall be automatically escalated to the responsible regulatory authority with **UNRESOLVED** status.
4. Where resolution requires deployment of a higher-capability validation node, the deployment and corroboration procedure shall follow the array-level fusion rules defined in Section 5, including eligibility criteria, inter-node calibration verification, and consistency checks.
5. Continue monitoring and archiving all other measurands for the affected station throughout the suspension period.

The escalation workflow, time bounds, notification recipients, and authority for manual override shall be documented in the system operating procedure and access-control policy. Any manual override of an **UNRESOLVED** result shall be recorded with the identity of the authorising operator, the justification, and any supporting evidence, and shall be retained in the audit archive.

Table 4: Measurand-Specific Compliance Decision Rules

Measurand		Default rule	Handling of UNCERTAIN result
Carrier Error	Frequency	Guard-Band	Trigger repeated measurement with $4\times$ extended averaging. If uncertain after 3 attempts, log as UNRESOLVED and escalate to operator.
Calibrated Power	Received	Shared-Risk	Flag for field-strength verification using a calibrated mobile unit or compliance-grade co-located node. Do not issue a violation notice based solely on an uncertain received-power result.
Field Strength		N/A Unsupported ^a	No compliance determination is made. Result shall be suppressed and flagged UNSUPPORTED in all reports.
Occupied Bandwidth		Guard-Band	Repeat with doubled FFT size (finer bin resolution). ^b If the uncertain zone width $2U < 5$ kHz after repetition, accept with a qualification note recording U , FFT size, and window function.
Adjacent-Channel Leakage		Guard-Band	Uncertain ACLR results shall not support a non-compliance determination. Escalate per the array-level coordination procedure in Section 5 using a higher-dynamic-range validation node.
Peak Deviation / Multiplex Power		Guard-Band	Repeat measurement with extended demodulation window. If uncertain after 3 attempts, escalate to operator. Do not issue violation notice on uncertain result alone.
Channel Occupancy		Guard-Band ^c	Report occupancy as a statistical estimate P_{occ} with a 95% confidence interval derived from the characterised false-alarm and missed-detection probabilities. Uncertain results below the confidence threshold shall not support a non-compliance determination.

^a Field Strength is Unsupported on the baseline HackRF One platform (Table 3). This row applies to upgraded platforms meeting the reclassification criteria.

^b Any change to FFT size shall be logged as a processing configuration change and is subject to version control per Table 3 precondition (1) for occupied bandwidth.

^c Channel Occupancy is inherently statistical. The Guard-Band rule is applied to the estimated occupancy fraction O_T against the policy-defined occupancy threshold, with U derived from the characterised false-alarm and missed-detection probabilities. The Simple Threshold rule shall not be used for Conditional Compliance-Grade occupancy reporting.